

Reading Cancer Referral Guidelines for Patients, and Testing Whether Routine Health Information Can Rank Cancer Risk

A guideline-based symptom and laboratory navigator and a pre-registered risk-ranking study in 37,915 US adults, with the bars set in advance and the failures reported

Palash Rakshit (author) · Clinical mentor: Dr. Rishikesh Chavan, UCI Health / CHOC (role and authorship to be agreed) · Oncovision project · Draft of 5 October 2026

Draft for mentor review. Not peer reviewed. The navigator applies the 2015 edition of UK guidance, its rules have **not** been reviewed by a clinician, and nothing here is tested on patients or is medical advice. The key findings are negative: five of eight pre-registered bars were missed.

Abstract

Background. Cancers are often found late because early symptoms and mildly abnormal blood tests are individually vague. Two ideas could help: apply published referral guidelines to a person's own symptoms and lab results in plain language, and rank people by cancer risk from information that is already routinely collected so that follow-up reaches those who need it first. We built the first, tested both, and set the pass-or-fail bars before looking.

Methods. The navigator is a rule engine, not a trained model: 34 rules transcribed from the 2015 edition of the UK NICE suspected-cancer guideline (NG12), each with its source page, using three-valued logic so that missing facts are requested rather than assumed. We ran it on 37,915 adults aged 40 and over from NHANES III and ten continuous NHANES cycles (1988-2018), linked to the National Death Index through 2019 (alert burden, cancer specificity among decedents, added value over age and sex, validity of its iron-deficiency assumption against measured ferritin, calibration of an age context layer). Separately, a risk-ranking study fitted five nested models for five-year cancer death on 1999-2006 (11,041 adults) and tested them on 2007-2014 (13,139 adults), from age and sex alone up to 380 variables covering every laboratory, examination and questionnaire item.

Results. Lab results alone alerted 12% to 17% of people aged 70-79 and 16% to 27% of those 80 and over, almost entirely through the guideline's anaemia-to-stool-test rule; no more than 2.7% in any band were told to see a doctor soon. **Five of the eight pass-or-fail bars were missed.** Among people who died, alerted people were not more likely to have died of cancer (adjusted odds ratio 0.74 (0.63 to 0.88) in 1999-2014; 1.18 (0.91 to 1.54) in NHANES III), and this held across sex, age group, follow-up window and a lag that removed the first two years. As a test for five-year cancer death the alert had sensitivity 16%, specificity 92% and positive predictive value 3.1% against a base rate of 1.5%. Adding it to age and sex did not improve prediction in the later era (AUC gain +0.005, 95% CI -0.009 to +0.019). In the risk-ranking study, age, sex, smoking and body mass index reached an AUC of 0.743 out of era and routine labs added -0.000; the 380-variable whole picture scored 0.629 (gain -0.114) and was badly miscalibrated (slope 0.087), and quadrupling the fitting data did not help. The age, sex and smoking context layer was well calibrated in the later era (slope 0.92). Patient-facing text averaged US grade

6.9. A companion lookup of US screening recommendations (breast, cervical, colorectal, lung, prostate) was built from the Task Force's own pages and tested at every age boundary.

Conclusions. A guideline-based navigator can be built, made readable and tested against bars set in advance. Its lab-driven alerts, and a model using everything the survey recorded, did not identify who would die of cancer better than age, sex, smoking and body mass index, which together reach AUC about 0.74. The honest uses are reading a guideline in plain language and putting an alert in proportion to what is ordinary for a person's age. The rules are UK guidance from 2015, have not been clinician-reviewed, and clinician validation is the necessary next step.

Keywords: cancer referral guidelines; symptom checker; laboratory medicine; early detection; NHANES; pre-registration; health literacy.

1. Introduction

A person who has had routine bloodwork usually learns one thing from it: which numbers fall outside a printed range. A blood count and a metabolic panel together produce about thirty values, and what matters in early cancer detection is rarely one of them being wrong. It is a small abnormality in a blood count, alongside a symptom the person has not thought to mention, alongside their age and smoking history. Doctors weigh these together. The person holding the report generally cannot.

Two bodies of work address parts of this. Clinical guidelines, most prominently the NICE guideline NG12 on suspected cancer, list the combinations of age, symptoms and test results that justify an urgent check, chosen so that the chance of cancer for someone meeting a criterion is about three percent or more.¹ Statistical risk models such as Qcancer combine symptoms, risk factors and blood tests into a probability.² Both are written for clinicians. Neither is designed to be handed to a person who has a lab report and a worry and wants to know whether it is worth asking about.

The second idea is screening in the sense of finding risk: using what is already known about a person to rank who is more likely to be affected, so that scarce follow-up goes to them first. Risk scores of this kind exist and some are validated, so the interest is not in the idea but in measuring honestly how much routine information adds once age is accounted for.

Multi-cancer detection from blood alone, in people with no symptoms, is a harder and still unresolved problem. Earlier work in this project built and tested several blood-only panels and withdrew most of them when they failed bars set in advance; the lesson was that routine blood values add little to age and sex for cancer outcomes in the general population. The route that remains well supported is narrower: people who already have a reason to be looking, such as a symptom, where guidelines already define who deserves a prompt check.

This paper describes a navigator for that case and, more importantly, measures it. The questions are practical ones. How many people would the lab rules alert if used without symptoms? Are the alerts specific to cancer, or do they mark ill health in general? Do they add to what age and sex already tell us? Is the tool's own simplifying assumption sound? Can an ordinary reader follow what it says? We committed

the questions and the pass-or-fail criteria to the repository before looking at results. Most of them failed, and we report that as the main finding.

2. Methods

2.1 Design

This is a development and retrospective evaluation study. No participant was contacted. All data are public, de-identified US survey records; the study is therefore not human-subjects research under the usual definitions, though the institution's own determination is being sought before any comprehension testing with readers (Section 4.6).

2.2 The rule engine

The navigator (`backend/navigator.py`) reads a request containing age, sex, smoking and asbestos history, a list of symptoms the person has ticked (with how often, where a rule needs it), lab values, and optional findings such as a positive stool test. It evaluates 34 rules held in a plain data file (`backend/navigator_rules.json`) that a clinician can read without reading code. The rules were transcribed from the full NICE NG12 guideline (2015 edition), and every rule records its source page. The engine contains no clinical wording of its own: the action text and the criteria shown to the user come from the file.

Three-valued logic. Every condition evaluates to true, false or unknown. A rule that could apply if one more fact were known is reported as "could apply if", naming exactly what is needed, and is not counted as a match. A missing smoking history or an unentered haemoglobin is therefore asked for and never assumed. Impossible inputs (such as a haemoglobin of 40 g/dL) are refused.

Alert levels. The guideline gives each recommendation a timeframe and a strength. These map to three levels: *see a doctor soon* (urgent or two-week criteria that are recommended), *worth raising* and *mention at a routine visit*. For a given cancer site only the highest level that applies is shown. The tool never states that a person has cancer, and a result of "no threshold met" is worded as different from "nothing is wrong".

How the navigator works

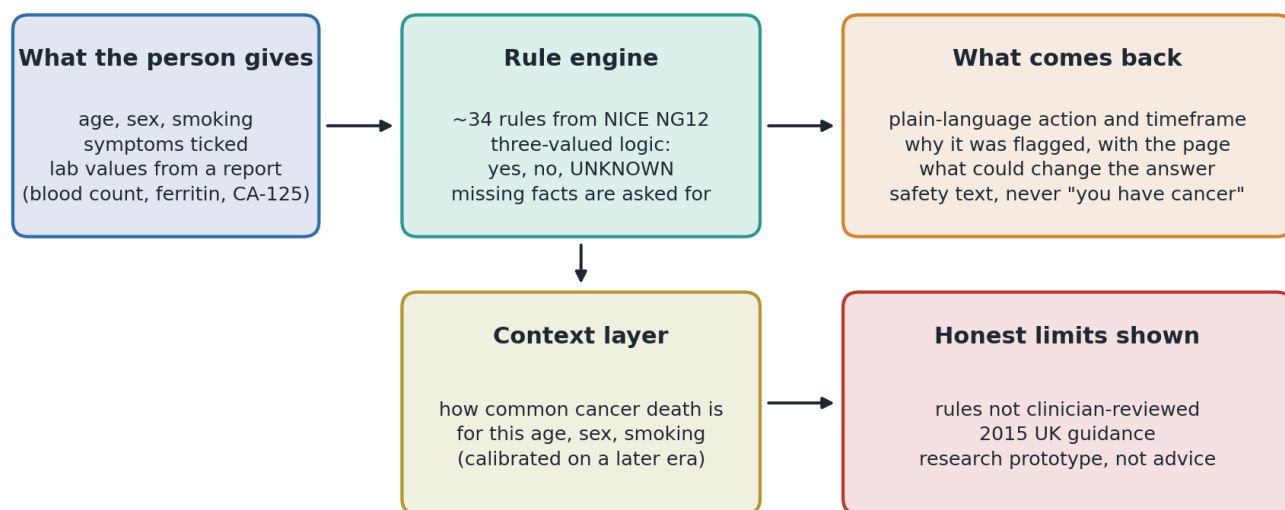


Figure 1. How the navigator works. Inputs on the left are optional; the rule engine asks for what it needs. The context layer and the limits are shown with every answer.

Table 1. The rules, by cancer site.

Cancer site	Rules	Alert levels it can produce
Anal cancer	1	worth raising
Bladder cancer	2	talk soon, mention
Bowel cancer	4	talk soon, worth raising
Breast cancer	3	talk soon, worth raising, mention
Kidney cancer	1	talk soon
Leukaemia	3	talk soon, worth raising
Lung cancer	3	talk soon, worth raising
Lymphoma	2	worth raising
Mesothelioma	1	talk soon
Oesophageal or stomach cancer	3	talk soon, mention
Ovarian cancer	4	worth raising
Pancreatic cancer	2	talk soon, worth raising
Prostate cancer	2	talk soon, mention
Several cancers	2	talk soon
Melanoma	1	talk soon (7-point checklist)

34 rules in total. Each carries its source page in the guideline. Levels come from the guideline's own timeframe and wording ('recommended' versus 'consider').

2.3 Laboratory findings

Raw lab values are converted into the findings the rules read (Table 2). Iron deficiency uses ferritin when it is entered. When it is not, the engine assumes iron deficiency from small red cells (MCV below 80 fL), and says so. Question 4 below tests how reasonable that assumption is.

Table 2. How lab values become the findings the rules read.

Derived finding	Definition used	Where it comes from
Anaemia	Haemoglobin below 13 g/dL (male) or 12 g/dL (female)	WHO definition for adults
Iron deficiency	Ferritin below 15 ng/mL. If no ferritin, MCV below 80 fL is used instead and the person is told so	Common laboratory cut-off; MCV rule is the tool's disclosed assumption
Iron-deficiency anaemia	Anaemia and iron deficiency together	Derived
Thrombocytosis	Platelets above $400 \times 10^9/L$	Guideline
Raised white count	White cells above $11 \times 10^9/L$	Common laboratory cut-off
Raised CA-125	35 U/mL or more	Guideline

2.4 Data

NHANES is a continuing US national survey with a physical examination and laboratory tests. NCHS links participants to the National Death Index, giving vital status and underlying cause of death through 31 December 2019. We used NHANES III and ten continuous cycles; cycles examined in 2015-2018 have under five years of follow-up for many people and so were used only for questions that need no outcome (Table 3). Participants who reported a previous cancer diagnosis were excluded where the questionnaire asked, and nobody was excluded for how they later died, a choice learned from an earlier cohort in this project that dropped everyone who died of another cause and thereby made every early death a cancer death. The outcome is **death from cancer** (NCHS underlying cause 2) within 60 months of the examination: a later and harsher endpoint than diagnosis, and one that cannot show whether an alert would have led to a correct referral.

Table 3. The data.

Source	Years	Adults 40+ with a blood count	Used for
NHANES III, linked to the National Death Index	1988-1994	9,335	Burden; cancer specificity; added value (external era)
Continuous NHANES, eight cycles with five-year follow-up	1999-2014	22,583	All questions; context layer
Continuous NHANES, two newest cycles	2015-2018	5,997	Burden and ferritin check only (follow-up too short)
Total	1988-2018	37,915	

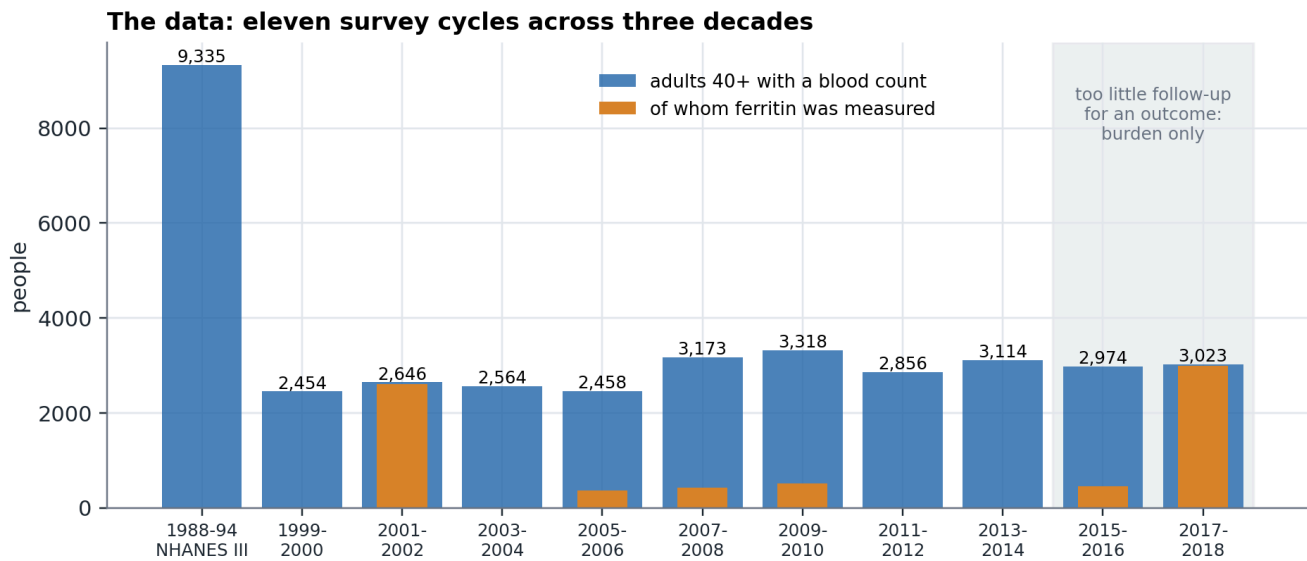


Figure 2. The data. Blue bars are adults aged 40 or over with a blood count. Orange shows how many also had ferritin measured, which differs by survey cycle. The shaded cycles have too little follow-up for an outcome.

2.5 What the engine was given

The real engine was run, not a re-implementation. Two versions were scored for every person. **Labs only** passed haemoglobin, MCV, platelets, white count and (where measured) ferritin, with no symptoms and no smoking history. **Labs plus proxies** added smoking (100 or more cigarettes in a lifetime) and three symptom stand-ins, passed only when present: *weight loss* (weight a year ago minus weight now of at least 5% of the earlier weight and not trying to lose weight, or at least 10% where that question was not asked), *chronic cough* (cough on most days for three months, asked 1999-2012) and *breathlessness* (asked of people 40 and over). NHANES never asked about bleeding, lumps, bowel change, appetite, tiredness or jaundice, so most of the guideline's symptoms are invisible here, and every alert rate below is a **lower bound** on what someone reporting those symptoms would see. The proxies are crude: weight loss in a survey includes illness and medication effects that a person ticking "no known reason" would exclude.

2.6 Pre-registered questions and bars

Six questions and their criteria were committed before analysis (docs/NAVIGATOR_EVIDENCE_PREREG.md) and are summarised with the results in Table 4. The earlier project had repeatedly found that results look better when the comparison is weak, so every prediction question uses age and sex as the baseline, and cancer specificity is tested only among people who died, so that "sick people die of things" cannot masquerade as a cancer signal.

2.7 Statistics

Proportions use Wilson 95% intervals.³ Cancer specificity among decedents (deaths within ten years of examination) used logistic regression of cancer versus other cause on the flag, age, age squared, sex and

examination year, with Wald intervals. Added value used logistic models of five-year cancer death with age, age squared and sex, with and without the alert, fitted on 1999-2006 and tested on 2007-2014 (and, exploratorily, fitted on 1999-2014 and tested on NHANES III); the AUC difference was bootstrapped 1,000 times, resampling cases and non-cases separately. Non-cancer deaths inside the window are counted as non-cases, which is a competing-risk simplification. All analyses are unweighted, because the aim was to exercise the engine on realistic profiles, not to estimate a national prevalence.

2.8 Context layer

To answer "how much of this is just my age", the tool reports how common five-year cancer death is among US adults of the same age, sex and smoking history: a logistic model of age, age squared, sex and smoking, fitted on 1999-2006 and tested on 2007-2014 (bar: calibration slope between 0.8 and 1.2), then refitted on all outcome cycles for use in the tool. It does not use labs or symptoms and is shown as a group average, not the person's risk.

2.9 Readability

Every sentence the tool can show was scored with the Flesch-Kincaid grade level.⁴ The bar, set before the first measurement, was a mean of at most 8 and no action line above 10. This is a floor and not a test of understanding.

2.10 Further descriptive and sensitivity analyses

To show whether the headline findings depend on a choice, the cancer-specificity analysis was repeated by sex, by age group (40-59, 60-74, 75 and over), by smoking status, in the subset with measured ferritin, with a five-year instead of ten-year window, and with the first 12 and 24 months of follow-up removed (to guard against illness already present at the examination). The alert was also treated as a diagnostic test for five-year cancer death (sensitivity, specificity, predictive values, positive likelihood ratio, each with Wilson intervals) and compared with simply flagging the same number of the oldest people. Absolute five-year cancer death was tabulated by age band for alerted and non-alerted people. None of these carries a pass-or-fail bar.

2.11 Risk-ranking study

The question was whether information beyond age predicts five-year cancer death. The risk-ranking cohort is NHANES 1999-2014 adults 40 and over with every laboratory, examination and questionnaire column recorded in at least six of eight cycles (over 300 columns, plus a missing-value indicator for each partly missing column). Models were fitted on cycles 1999-2006 and tested on cycles 2007-2014, which differ in population, laboratory methods and questionnaire. Five feature sets were compared: age and sex (S0); plus smoking and body mass index (S1, the strong baseline); plus 22 routine blood values (S2); all 380 variables (S3); and, post hoc, the same without the missing-value indicators (S3b). For each set, ridge logistic regression or gradient-boosted trees were chosen by five-fold cross-validation inside the fitting era; hyperparameters were fixed in advance. Bars were committed beforehand (docs/RISK_RANKING_PREREG.md): R1, an AUC gain of at least 0.02 for S3 over S1 with a bootstrap interval above zero; R2, a better separation of cancer from other deaths among people who died within five

years; R3, a calibration slope between 0.8 and 1.2. R4 (learning curve on 25%, 50% and 100% of the fitting data) and R5 (share of cancer deaths in the top-scored 10%, 20% and 30%) have no bar.

2.12 The screening lookup (companion tool)

A second page, `/screening`, answers a different question: which routine screenings do US guidelines recommend for a person at average risk? It is a lookup, not a prediction. It applies the five US Preventive Services Task Force (USPSTF) recommendations for breast, cervical, colorectal, lung and prostate cancer to age, sex at birth, whether the cervix is present, and smoking history (smoking now, years since quitting, pack-years). The age limits, tests, intervals and grades were read from the Task Force's own recommendation pages on 5 October 2026 and are recorded with each item. Every boundary (for example age 39 versus 40 for mammography, 15 versus 15.5 years since quitting for lung screening, 19.9 versus 20 pack-years) has an automated test written from the published wording. Missing facts are asked for, and the tool states that it covers average-risk adults only and that it does not say other cancers cannot be screened for. Prostate testing is never called "due", because the Task Force's grade for ages 55 to 69 is an individual decision.

2.13 Software and reproducibility

Backend: Python and FastAPI; frontend: Next.js, deployed publicly. There are 120 automated tests, including properties such as "a symptom-free person with normal labs is never told to see a doctor soon over 300 random profiles", "the tool never says 'you have cancer'" and "a rule that refers to a symptom the form cannot ask about is a loud failure". Every number in this paper is regenerated by scripts in the repository, and the tables and figures are built directly from their output. Reporting follows the spirit of TRIPOD+AI where it applies;⁵ because the engine is not a trained model there is no training-set leakage to guard against in the rules themselves, only in the context layer, which is validated out of era.

3. Results

3.1 Summary against the pre-registered bars

Table 4. Pre-registered questions and what happened.

#	Pre-registered question and bar	Result	Outcome
1	Burden is stable across eras: 'see a doctor soon' at most 3% and any alert at most 20% in every age band and era	Worst 'soon' 2.7%; worst any-alert 26.7% (age 80+, 2015-2018)	Missed (any-alert, ages 80+)
2	Alerted people are more likely to die of cancer than of something else, among decedents (adjusted OR interval above 1 in 1999-2014, and OR above 1 in NHANES III)	1999-2014: OR 0.74 (0.63 to 0.88). NHANES III: OR 1.18 (0.91 to 1.54)	Missed
3	An alert adds to age and sex for five-year cancer death in the held-out era (AUC-gain interval excludes 0)	Labs and proxy symptoms: +0.005 (-0.009 to +0.019); labs only +0.000	Missed
4	Report how well MCV below 80 stands in for ferritin (no bar)	Sensitivity 65%, PPV 71%	Reported
5	Which rules fire (no bar)	Colorectal rules account for almost all lab-driven alerts	Reported
6	Age, sex and smoking context layer is calibrated in a later era (slope 0.8 to 1.2)	Slope 0.92; AUC 0.747	Met
R1	Risk ranking: the whole picture (380 variables) beats age, sex, smoking and BMI by at least 0.02 AUC, out of era	Whole picture 0.629 vs 0.743: gain -0.114 (-0.161 to -0.069)	Missed
R2	Risk ranking: among decedents, the whole picture separates cancer from other deaths better than the baseline (CI above 0)	Decedent AUC 0.53 vs 0.47: gain +0.063 (+0.005 to +0.122)	Met , but read with care (Section 3.10)
R3	Risk ranking: whole-picture calibration slope 0.8 to 1.2 out of era	Slope 0.087 (baseline: 0.978)	Missed
F1	Follow-up: a curated, cycle-stable 29-variable set beats the baseline by 0.01 AUC with interval above 0, calibrated, and better among decedents	Gain +0.009 (-0.007 to +0.026); slope 0.987	Missed
R4, R5	Does more data help; triage value (no bars)	More data did not help; whole picture found fewer cancer deaths in its top 20% than the baseline	Reported

Bars were committed to the repository before each analysis was run (docs/NAVIGATOR_EVIDENCE_PREREG.md and docs/RISK_RANKING_PREREG.md).

3.2 Who is in the analysis

The navigator cohort comprises 28,580 adults aged 40 and over from ten NHANES cycles, of whom 22,583 (examined 1999-2013) have enough follow-up for a five-year outcome, and 9,335 from NHANES III. Table 5 describes the three continuous eras. They are similar in age and sex and differ in smoking

(falling from 52% to 44%), mean haemoglobin and the share with a lab alert. The five-year cancer death rate was 1.5% overall.

Table 5. Who is in the analysis, by survey era.

Characteristic	1999-2006	2007-2014	2015-2018
Adults 40+ with a blood count	10,122	12,461	5,997
Mean age (years)	59.8	58.4	59.0
Female	50%	52%	52%
Ever smoked	52%	48%	44%
Mean haemoglobin (g/dL)	14.28	13.99	13.91
Mean MCV (fL)	90.6	89.6	88.9
Mean platelets ($\times 10^9/L$)	265	242	237
Mean white count ($\times 10^9/L$)	7.14	7.1	7.2
Weight-loss proxy present	11%	11%	13%
Died by end of 2019	36%	16%	4%
Cancer deaths (any time to 2019)	705	410	40
Any lab-only alert	6.7%	8.6%	9.5%

Later cycles have fewer deaths because they have had less time to die, not because they were healthier.

Table 3. The data.

Source	Years	Adults 40+ with a blood count	Used for
NHANES III, linked to the National Death Index	1988-1994	9,335	Burden; cancer specificity; added value (external era)
Continuous NHANES, eight cycles with five-year follow-up	1999-2014	22,583	All questions; context layer
Continuous NHANES, two newest cycles	2015-2018	5,997	Burden and ferritin check only (follow-up too short)
Total	1988-2018	37,915	

NHANES: US National Health and Nutrition Examination Survey. Mortality follow-up through 31 December 2019 from the NCHS public-use linked mortality files.

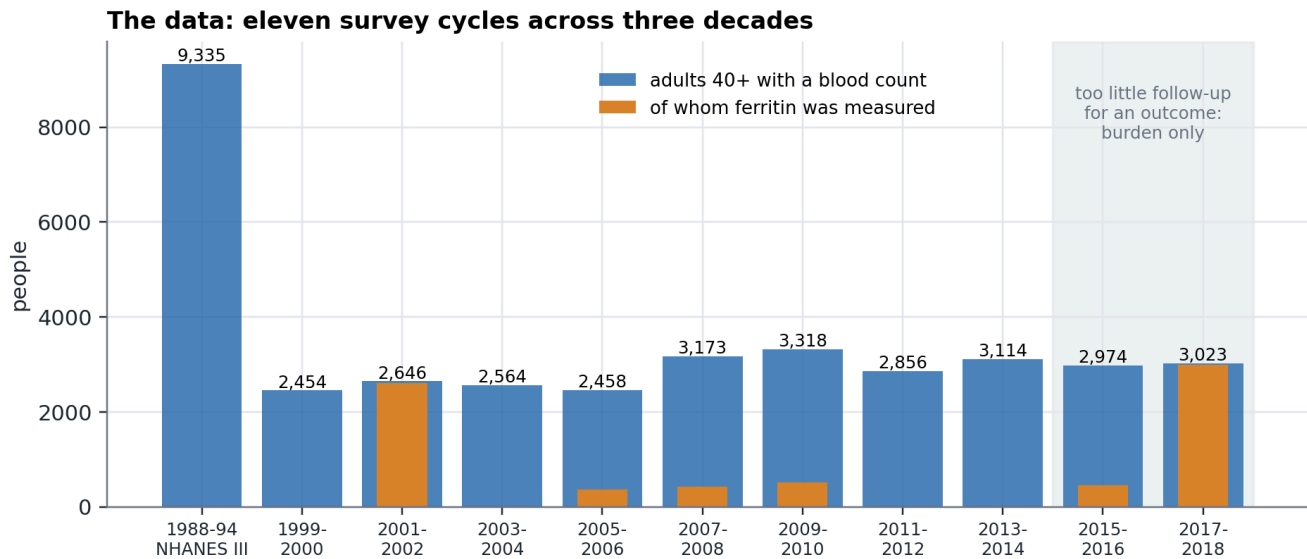


Figure 2. The data. Blue bars are adults aged 40 or over with a blood count. Orange shows how many also had ferritin measured, which differs by survey cycle. The shaded cycles have too little follow-up for an outcome.

3.3 Alert burden

With lab results alone and no symptoms, the share of adults alerted rises steeply with age (Table 6, Figure 3): about 4% to 8% at 40-49, 1% to 2% at 50-59, and 16% to 27% at 80 and over. The pattern is close to identical in all four eras, spanning thirty years of laboratory methods. "See a doctor soon" stayed at or below 2.7% in every band and era, and is zero below age 60.

Almost every lab-driven alert comes from one guideline rule: anaemia in people 60 or over, or iron-deficiency anaemia under 60, leads to a stool test. In the two newest cycles, where ferritin was measured for most people, this rule alerted 13.7% of women but 1.6% of men aged 40-49, a difference consistent with menstrual iron loss, which the rule does not take into account. That is a real feature of the guideline applied to people who may have a benign explanation. The pre-registered 20% line for any alert was crossed only at age 80 and over, in three of the four eras (21%, 23% and 27%).

How many symptom-free adults the lab rules alert, by age and era

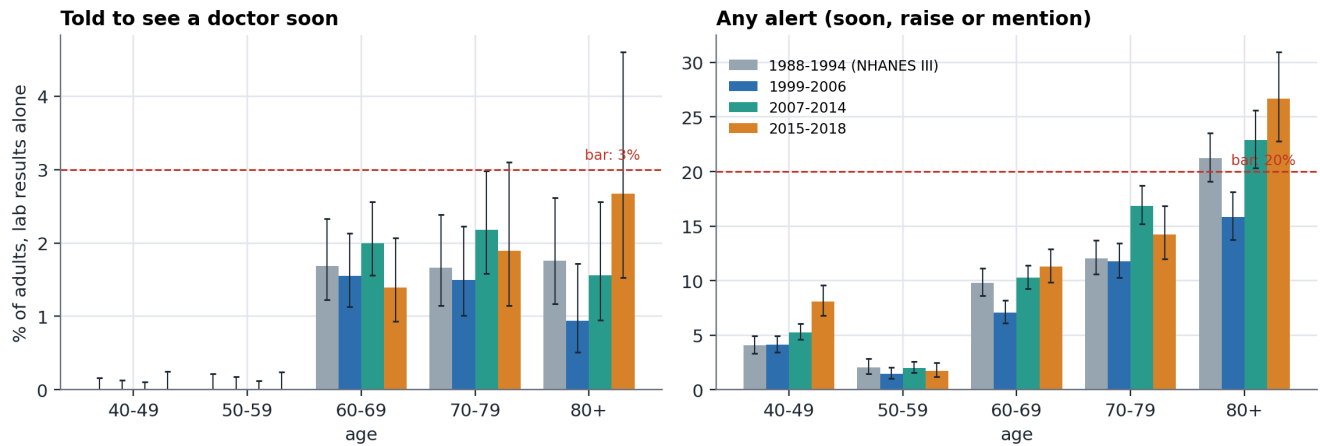


Figure 3. Share of symptom-free adults alerted by lab results alone, by age and era. Red dashed lines are the pre-registered bars. Error bars are 95% Wilson intervals.

Table 6. Share of adults alerted by lab results alone, by era and age.

Era	Age	Adults	See a doctor soon	Any alert (95% interval)
1988-1994 (NHANES III)	40-49	2,414	0.0%	4.1% (3.3 to 4.9)
	50-59	1,746	0.0%	2.1% (1.5 to 2.8)
	60-69	2,183	1.7%	9.8% (8.6 to 11.1)
	70-79	1,683	1.7%	12.1% (10.6 to 13.7)
	80+	1,309	1.8%	21.2% (19.1 to 23.5)
1999-2006	40-49	2,932	0.0%	4.1% (3.5 to 4.9)
	50-59	2,138	0.0%	1.5% (1.0 to 2.1)
	60-69	2,388	1.6%	7.1% (6.1 to 8.2)
	70-79	1,597	1.5%	11.8% (10.3 to 13.5)
	80+	1,067	0.9%	15.8% (13.8 to 18.1)
2007-2014	40-49	3,565	0.0%	5.3% (4.6 to 6.1)
	50-59	3,187	0.0%	2.0% (1.6 to 2.6)
	60-69	3,046	2.0%	10.3% (9.2 to 11.4)
	70-79	1,701	2.2%	16.9% (15.2 to 18.7)
	80+	962	1.6%	22.9% (20.3 to 25.6)
2015-2018	40-49	1,522	0.0%	8.1% (6.8 to 9.6)
	50-59	1,575	0.0%	1.7% (1.2 to 2.5)
	60-69	1,657	1.4%	11.3% (9.8 to 12.9)
	70-79	793	1.9%	14.2% (12.0 to 16.9)
	80+	450	2.7%	26.7% (22.8 to 30.9)

3.4 Are the alerts specific to cancer?

No. Among the 3,864 people in 1999-2014 who died within ten years, those alerted by labs alone were *less* likely to have died of cancer than of something else (13.5% versus 21.7%; adjusted odds ratio 0.63 (0.49 to 0.80)). Adding the proxy symptoms moved the estimate to 0.74 (0.63 to 0.88). The result did not replicate in direction in NHANES III (1.18 (0.91 to 1.54), interval including 1), and the two estimates differ more than chance would explain ($z = -3.473$, $p = 0.0005$). So the size of the effect is not stable across eras. What is stable is that in no era did an alert identify cancer deaths.

The direction held under every other choice we tried (Table 8, Figure 5): a five-year window, removing the first one or two years of follow-up, women and men separately, three age groups, smokers and non-smokers, and the subset with measured ferritin. Women showed the strongest effect (OR 0.41 (0.26 to 0.64)), and the ferritin subset the widest interval. No subgroup pointed toward cancer. The upper bound crossed 1 only in subgroups with few alerted decedents.

Looking at one flag at a time (Table 11, Figure 4; exploratory), anaemia was the clearest: among decedents, anaemic people had about 40% lower odds of having died of cancer rather than another cause (OR 0.57 (0.45 to 0.72)). The proxies for weight loss and breathlessness pointed the same way. Smoking, the one flag that is not a symptom or lab, pointed the expected way (OR 1.30 (1.09 to 1.56)).

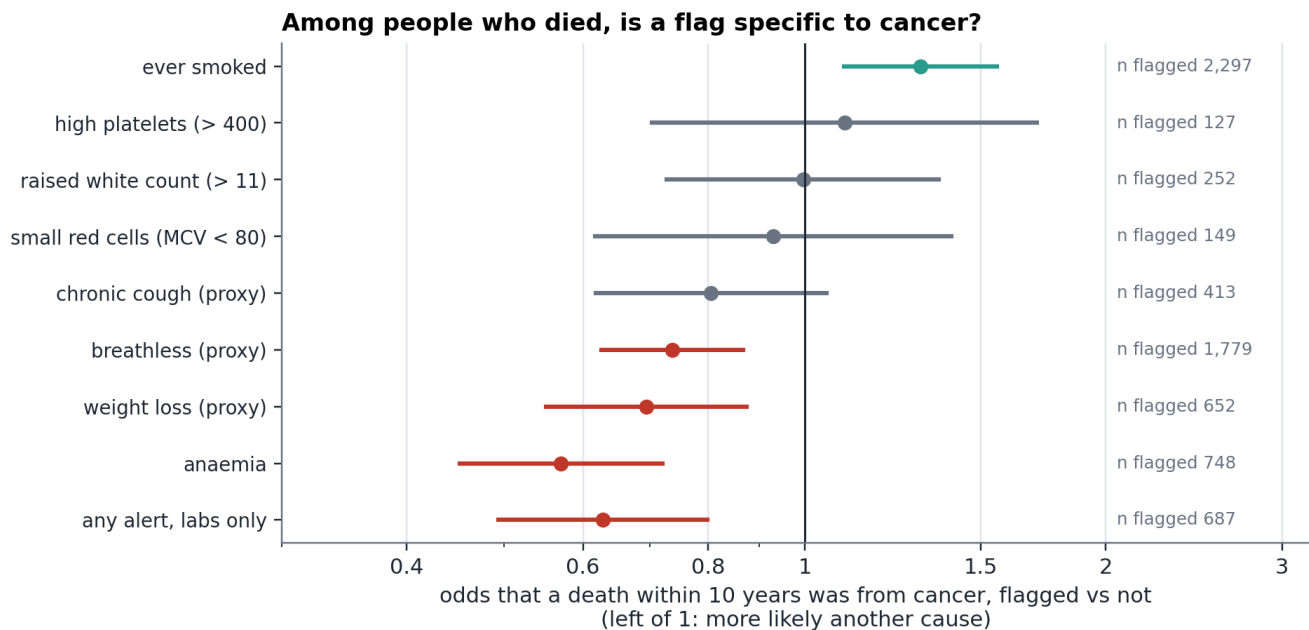


Figure 4. Among people who died within ten years, the odds that the death was from cancer rather than another cause, for people with each flag versus those without. Red intervals lie entirely below 1. Adjusted for age, sex and year.

Table 7. Cancer specificity among people who died within ten years.

Group	Deaths in 10 years	Alerted	Cancer share, alerted	Cancer share, not alerted	Adjusted OR (95% CI)
1999-2014 labs only	3,864	687	13.5%	21.7%	0.63 (0.49 to 0.80)
1999-2014 labs and proxies	3,864	2,094	18.5%	22.3%	0.74 (0.63 to 0.88)
NHANES III labs only	2,324	400	23.0%	22.4%	1.18 (0.91 to 1.54)

Decedents only: everyone here died within ten years, so the question is whether the death was from cancer. Adjusted for age, age squared, sex and (pooled) examination year.

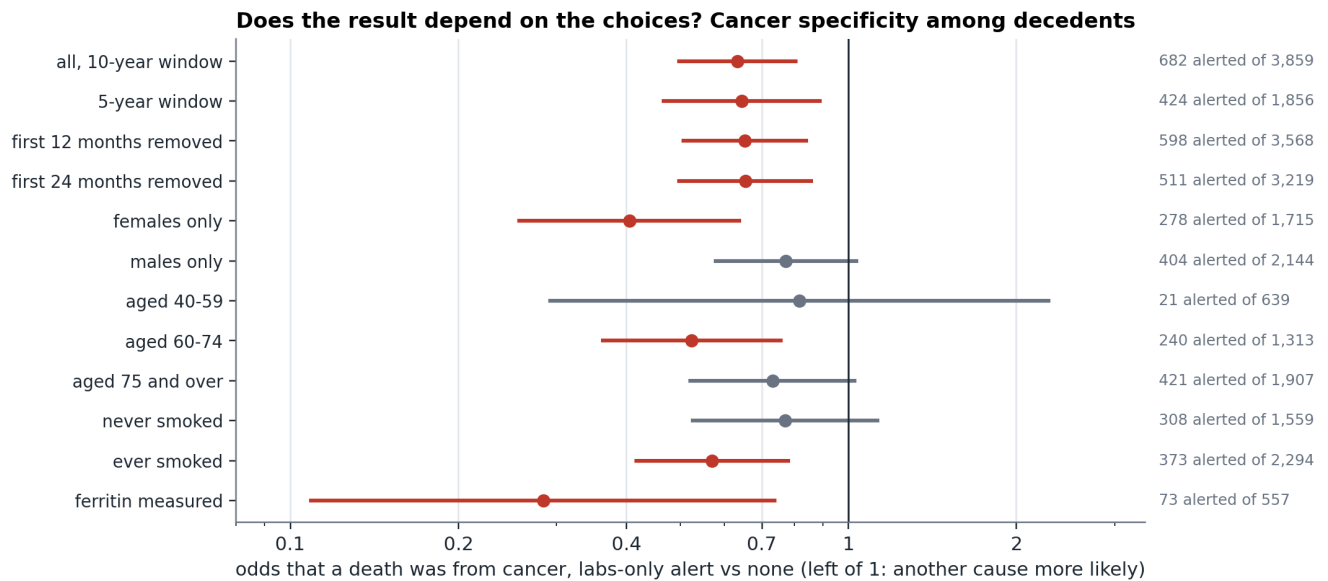


Figure 5. The result under twelve different choices about who is included. Red intervals lie entirely below 1. No choice moves the estimate toward cancer.

Table 8. Sensitivity analyses for cancer specificity.

Analysis	Decedents	Alerted	Adjusted OR (95% CI)
all, 10-year window	3,859	682	0.63 (0.50 to 0.81)
5-year window	1,856	424	0.64 (0.47 to 0.89)
first 12 months removed	3,568	598	0.65 (0.51 to 0.84)
first 24 months removed	3,219	511	0.65 (0.50 to 0.86)
females only	1,715	278	0.41 (0.26 to 0.64)
males only	2,144	404	0.77 (0.58 to 1.03)
aged 40-59	639	21	0.82 (0.29 to 2.28)
aged 60-74	1,313	240	0.52 (0.36 to 0.76)
aged 75 and over	1,907	421	0.73 (0.52 to 1.03)
ferritin measured	557	73	0.28 (0.11 to 0.74)
never smoked	1,559	308	0.77 (0.53 to 1.13)
ever smoked	2,294	373	0.57 (0.42 to 0.78)

Labs-only alert, people who died. Every row is a different choice about who to include; none changes the direction.

Table 11. Each flag on its own, among decedents (exploratory).

Single flag	Decedents flagged	Adjusted OR for dying of cancer rather than another cause (95% CI)
anaemia	748 of 3,864	0.57 (0.45 to 0.72)
small red cells (MCV < 80)	149 of 3,864	0.93 (0.62 to 1.40)
high platelets (> 400)	127 of 3,864	1.10 (0.70 to 1.71)
raised white count (> 11)	252 of 3,864	1.00 (0.73 to 1.36)
weight loss (proxy)	652 of 3,864	0.69 (0.55 to 0.87)
chronic cough (proxy)	413 of 3,864	0.81 (0.62 to 1.05)
breathless (proxy)	1,779 of 3,864	0.74 (0.63 to 0.87)
ever smoked	2,297 of 3,864	1.30 (1.09 to 1.56)
any alert, labs only	687 of 3,864	0.63 (0.49 to 0.80)

Exploratory. Nine comparisons; only intervals well away from 1 should be believed. 'Proxy' means a crude NHANES stand-in for a symptom (see Methods).

3.5 The alert as a test, and absolute risk

Treated as a screening test for five-year cancer death (Table 9), the labs-only alert flagged 7.8% of adults, found 16% of those who died of cancer, and had a positive predictive value of 3.1% (a positive likelihood ratio of 2.08) against a base rate of 1.5%. So it lifted the chance from 1.5% to 3.1%, an improvement but a small one in absolute terms, and one that mostly reflects age: flagging the same number of the oldest

people by age alone gave a PPV of 4.5% and sensitivity of 23%. Including the proxy symptoms flagged 35% of adults at a PPV of 2.3%.

The absolute numbers are reassuring in every group (Table 10, Figure 6). From age 50 the alerted group had a higher five-year cancer death rate than the non-alerted group (rate ratios 1.3 to 2.2), but the intervals overlap and even among alerted people aged 80 and over about 94 in 100 did not die of cancer within five years. At 40-49 there was no difference. This is consistent with the alert marking general ill health somewhat more than cancer: raising the rate of cancer death modestly while raising the rate of all death more.

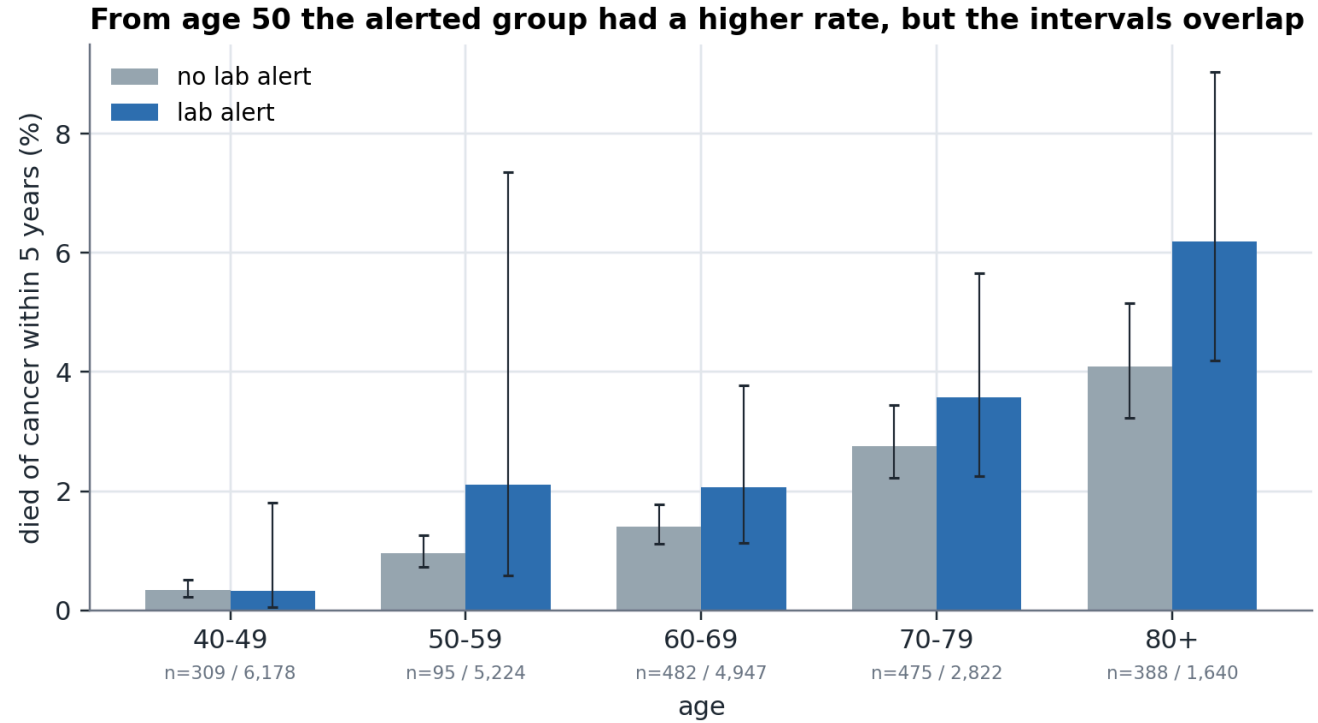


Figure 6. Five-year cancer death by age band for people with and without a labs-only alert. Error bars are 95% Wilson intervals; group sizes are under each band.

Table 9. The lab alert treated as a test for five-year cancer death.

Rule for who is flagged	Share flagged	Sensitivity (95% CI)	Specificity	PPV (95% CI)	LR+
any alert, labs only	7.8%	15.9% (12.4 to 20.2)	92.4%	3.1% (2.4 to 4.0)	2.08
see a doctor soon, labs only	0.8%	1.8% (0.8 to 3.8)	99.2%	3.3% (1.5 to 6.9)	2.2
any alert, labs + proxy symptoms	34.9%	53.2% (47.9 to 58.5)	65.4%	2.3% (2.0 to 2.7)	1.54
any alert, labs only (versus oldest 1,749 by age alone)	7.8%	22.9% (18.8 to 27.7)	92.5%	4.5% (3.6 to 5.5)	3.05
any alert, labs + proxy symptoms (versus oldest 7,869 by age alone)	34.9%	67.3% (62.2 to 72.1)	65.6%	2.9% (2.6 to 3.3)	1.96

Five-year cancer death, examinations 1999-2013. PPV: of those flagged, the share who died of cancer within five years. The base rate is 1.5%. The age-only rows flag the same number of people but choose the oldest, as a fair comparison.

Table 10. Absolute five-year cancer death by age, alerted versus not.

Age	Alerted: n	Died of cancer in 5 years	Not alerted: n	Died of cancer in 5 years	Rate ratio
40-49	309	0.32% (0.06 to 1.81)	6,178	0.34% (0.22 to 0.52)	0.94
50-59	95	2.11% (0.58 to 7.35)	5,224	0.96% (0.73 to 1.26)	2.2
60-69	482	2.07% (1.13 to 3.78)	4,947	1.41% (1.12 to 1.78)	1.47
70-79	475	3.58% (2.25 to 5.66)	2,822	2.76% (2.22 to 3.44)	1.3
80-+	388	6.19% (4.19 to 9.04)	1,640	4.09% (3.23 to 5.16)	1.51

Labs-only alert. Absolute risk is low in every group, including the alerted: even in the oldest band, about 94 in 100 alerted people did not die of cancer within five years.

3.6 Do alerts add to age and sex?

Age and sex alone predicted five-year cancer death with AUC 0.73 in the held-out era. Adding the labs-only alert changed this by +0.000 (95% CI -0.005 to +0.005); adding the alert from labs and proxy symptoms changed it by +0.005 (-0.009 to +0.019). Both intervals include zero, so the bar was missed. In the exploratory NHANES III test the gain was +0.009 (+0.005 to +0.014), statistically above zero but under one point of AUC and not in the pre-registered test.

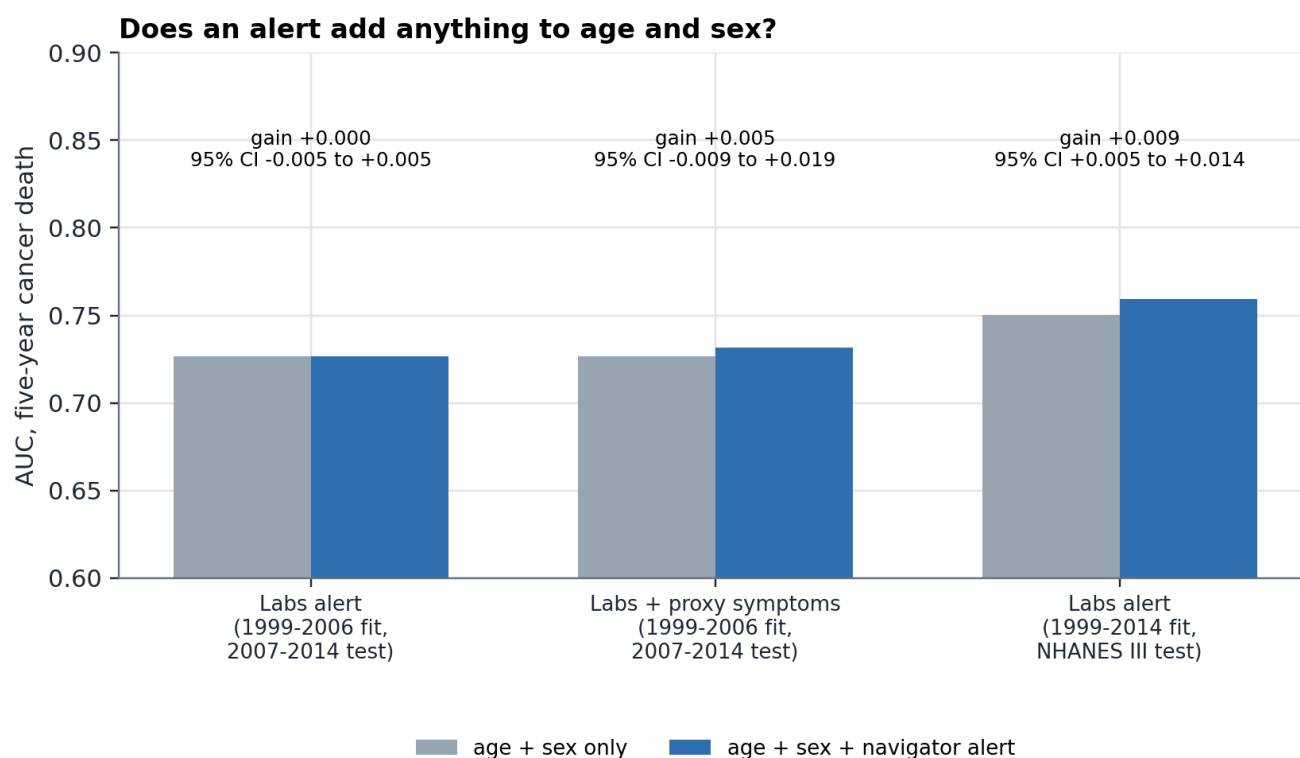


Figure 7. AUC for five-year cancer death with and without the navigator alert. Age and sex do nearly all the work.

Table 12. Does the alert add to age and sex for five-year cancer death?

Comparison	Test people	Cancer deaths	AUC, age + sex	AUC, with alert	Gain (95% CI)
Labs alert; fit 1999-2006, test 2007-2014	12,438	179	0.727	0.727	+0.000 (-0.005 to +0.005)
Labs + proxy symptoms; fit 1999-2006, test 2007-2014	12,438	179	0.727	0.732	+0.005 (-0.009 to +0.019)
Labs alert; fit 1999-2014, test NHANES III (exploratory)	9,326	238	0.750	0.759	+0.009 (+0.005 to +0.014)

AUC: probability that a person who died of cancer within five years has a higher score than one who did not. Bootstrap intervals, 1,000 resamples.

3.7 The tool's own iron-deficiency assumption

Among 7,373 adults with a measured ferritin, 817 were anaemic and 328 of those were iron-deficient by ferritin. Using MCV below 80 as a stand-in caught 65% of them and was right 71% of the time. Among 2,743 adults 60 or over with a ferritin, the proxy told 53 to see a doctor soon where measured ferritin told 36. A person who enters ferritin gets a more accurate answer than one who does not, and the tool says which one they got.

Testing the tool's own disclosed assumption

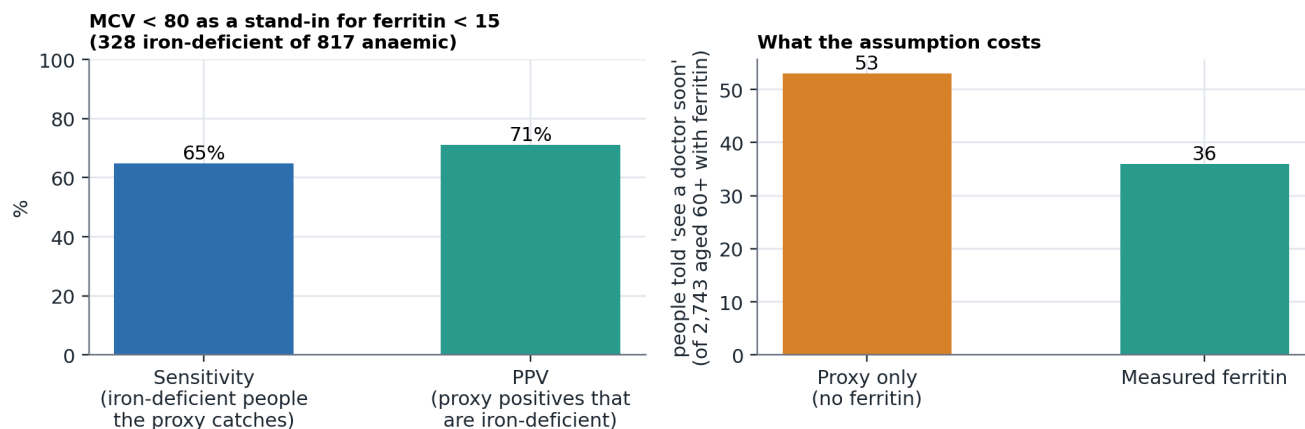


Figure 8. Left: how well MCV below 80 stands in for ferritin below 15 among anaemic adults. Right: the cost in alerts for people aged 60 and over.

3.8 Which rules fire

Table 13. Rules that fired, labs plus proxy symptoms.

Rule	People it fired for	Died of cancer within 5 years	Mean age
Chest X-ray (lung)	5,953	2.3%	60
Stool test for hidden blood (bowel)	3,175	2.6%	66
Unexplained weight loss, several cancers	2,520	1.8%	60
Consider ovarian tests	1,446	1.2%	60
Urgent bowel referral	184	3.3%	70
Consider chest X-ray (lung)	127	0.8%	58
Consider endoscopy (stomach, gullet)	55	3.6%	69
Everyone in the analysis	22,560	1.5%	

Labs plus proxy symptoms, 1999-2014. Descriptive only: the groups differ in age, and a higher rate in a group is mostly its older age.

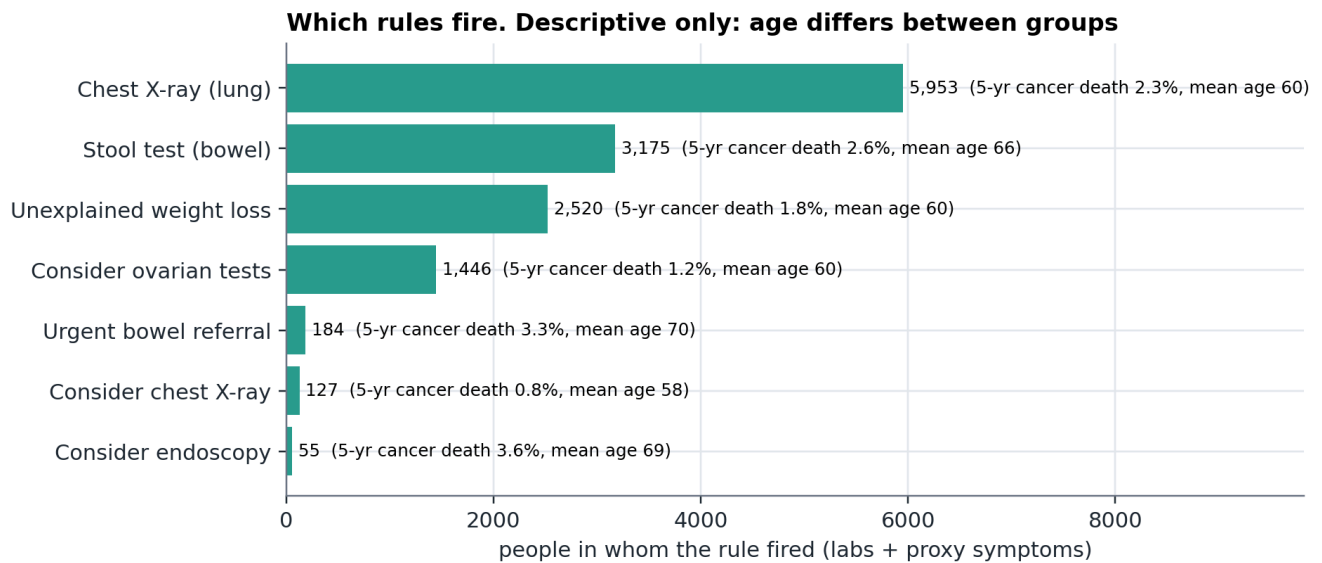


Figure 9. Which rules fire for labs plus proxy symptoms. The lung rule fires most often because breathlessness and smoking are common in older NHANES adults, not because of an excess of cancer.

3.9 The age context layer

Fitted on 1999-2006 and tested on 2007-2014, the age, sex and smoking model had AUC 0.747 and a calibration slope of 0.92, inside the pre-registered 0.8 to 1.2. Its Brier score (0.01398) was slightly better than predicting the average for everyone (0.01419). The model was refitted on 24,180 adults with 379 five-year cancer deaths for use in the tool. Typical outputs range from about 2 in 1,000 for a 45-year-old woman who has never smoked to about 50 in 1,000 for a 75-year-old man who has smoked.

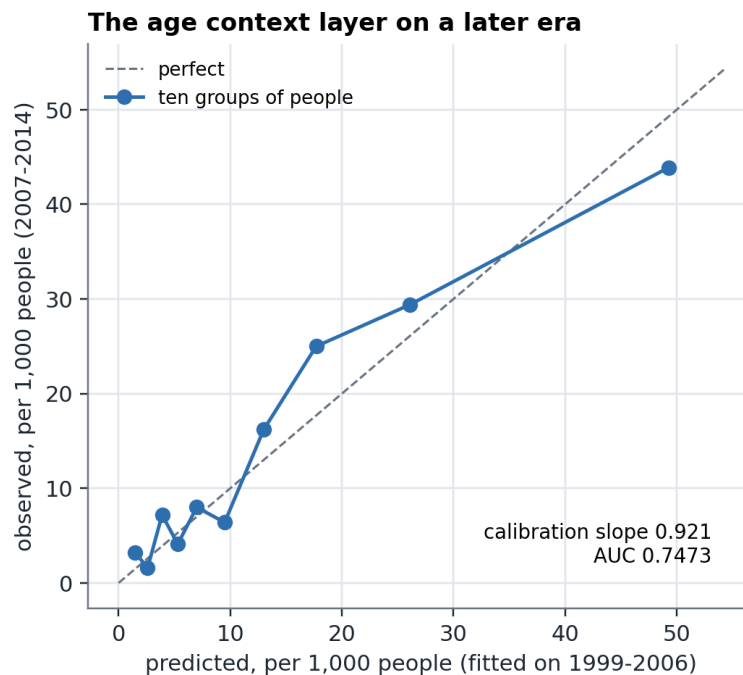


Figure 10. Calibration of the context layer in the later era.

Table 14. Calibration of the context layer by tenth.

Predicted per 1,000	Observed per 1,000	People
1.5	3.2	1,250
2.6	1.6	1,268
3.9	7.2	1,254
5.3	4.1	1,208
7.0	8.0	1,255
9.5	6.4	1,249
13.0	16.2	1,298
17.7	25.0	1,202
26.1	29.4	1,225
49.3	43.9	1,229

Ten equal-sized groups of the 2007-2014 people, ordered by predicted risk, from a model fitted only on 1999-2006.

3.10 Risk ranking: does everything a survey knows beat the obvious baseline?

Table 15 and Figure 11 show the main result. Age and sex alone ranked five-year cancer death with AUC 0.720 in the held-out era. Adding smoking and body mass index raised this to 0.743 (gain +0.024, 95% CI +0.007 to +0.042), which was the only step that helped. Adding 22 routine blood values changed it by -0.000 (-0.021 to +0.018), which is nothing. This repeats, on a larger sample and a later era, the project's earlier finding that routine blood values add little to age and sex for cancer outcomes.

The 380-variable whole picture did worse: AUC 0.629 against the baseline's 0.743 (gain -0.114, -0.161 to -0.069), and a calibration slope of 0.087, meaning its predictions barely tracked outcomes. Bars R1 and R3 were missed. Inside the fitting era, cross-validation gave this model 0.737, so the failure is specific to moving to a later era. The permutation importances point to why (Table 19): most of the largest entries are missing-value flags, that is, whether a questionnaire item was asked at all, and that depends on the survey cycle, which also differs in follow-up and baseline mortality. A model can learn "cycle" from those flags and look good by cross-validation inside the era, then fail on a new one. Removing the flags (a post hoc analysis, not pre-registered) restored the AUC to 0.716 but did not beat the baseline (gain -0.028, -0.061 to +0.006). The explanation was then tested with a follow-up written down in advance (Table 20): the missing-value flags alone predict the survey era with AUC 0.93, and the same whole-picture model scores 0.75 when fitted and tested on random halves of all cycles against 0.63 across eras. That is what an artefact of survey cycle would produce, though it does not prove it is the only cause.

We then tried the obvious repair: a curated set of 29 variables a clinician would consider relevant (weight change, HbA1c, albumin, diabetes, heart, lung and liver conditions and others), all present in every cycle, with no missing-value flags. It reached AUC 0.752 against the baseline's 0.743, a gain of +0.009 (95% CI -0.007 to +0.026), and was well calibrated (slope 0.987). The gain was too small and too uncertain to meet the bar, so a model with more clinically sensible variables is no better than four numbers in this survey.

Bar R2 was met: among the 1,007 people who died within five years in the later era (200 of cancer), the whole picture separated cancer deaths from other deaths better than the baseline did (AUC 0.53 against 0.47). This should not be read as the whole picture being cancer-specific: 0.53 is barely above chance, and the baseline's 0.47 is below it because age and smoking point toward non-cancer deaths among decedents, so the comparison is against a baseline that does worse than coin-flipping at this particular task. The whole picture is also the model that failed R1 and R3.

More data did not help (Table 18, Figure 13): fitting on 25%, 50% and 100% of the fitting era left the whole picture between 0.62 and 0.66, while the four-variable baseline stayed near 0.74. The problem is not the sample size in this range; it is that the extra variables carry era-specific patterns and little cancer signal. For triage (Table 17, Figure 12), the highest-scored 20% of people under the baseline contained 50% of five-year cancer deaths, against 46% under age and sex alone and 40% under the whole picture, where chance would give 20%.

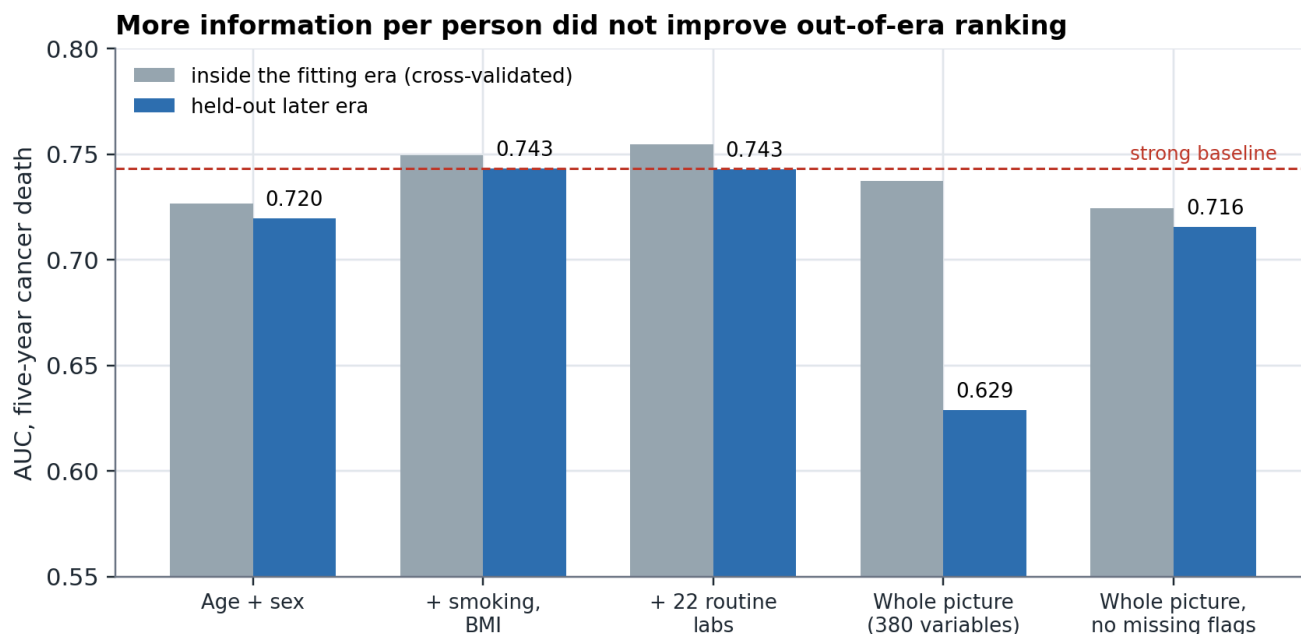


Figure 11. AUC for five-year cancer death by how much information the model sees. Grey: cross-validated inside the fitting era. Blue: the later era the model never saw. Red line: the strong baseline.

Table 15. Risk ranking: more information per person, fitted on one era and tested on another.

Feature set	Variables	Model chosen	AUC inside fitting era (cross-validated)	AUC in held-out later era
So age+sex	2	ridge	0.727	0.720
S1 +smoking, BMI	4	ridge	0.750	0.743
S2 +22 routine labs	65	ridge	0.755	0.743
S3 whole picture	380	ridge	0.737	0.629
S3b whole picture, no missing flags	206	ridge	0.724	0.716

Fitted on 1999-2006 (11,041 adults, 179 cancer deaths); tested on 2007-2014 (13,139 adults, 200 cancer deaths). The model type was chosen by cross-validation inside the fitting era, never on the test era.

Table 16. What each addition buys, in the held-out era.

Comparison, held-out era	AUC gain	95% CI
So age+sex -> S1 +smoking, BMI	+0.024	+0.007 to +0.042
S1 +smoking, BMI -> S2 +22 routine labs	-0.000	-0.021 to +0.018
S2 +22 routine labs -> S3 whole picture	-0.114	-0.159 to -0.072
Whole picture, no missing flags (post hoc) vs age + sex + smoking + BMI	-0.028	-0.061 to +0.006
Whole picture vs age + sex + smoking + BMI, among 1,007 people who died within five years (cancer vs other cause)	+0.063	+0.005 to +0.122

Bootstrap intervals. Each row adds information to the one before it. Note the last row: among decedents the age-and-smoking baseline has AUC 0.47 (below chance: older, smoking decedents are more often non-cancer deaths) and the whole picture 0.53, barely above chance.

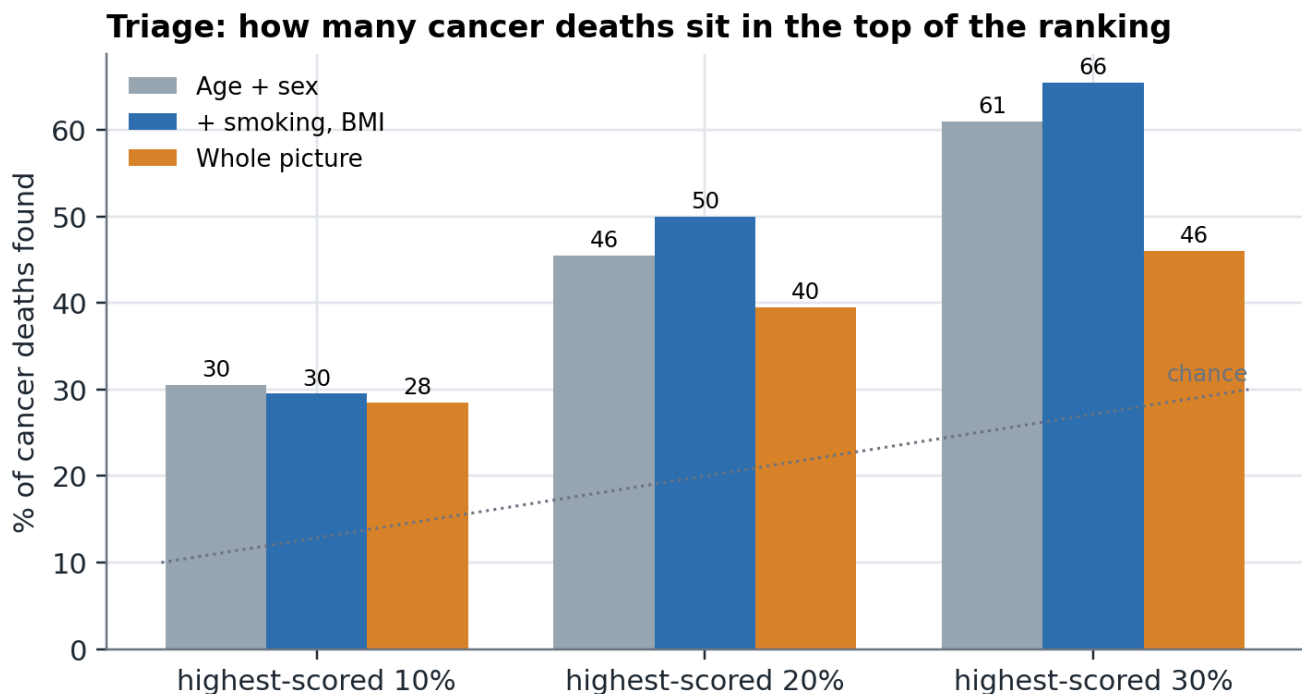


Figure 12. Share of five-year cancer deaths found in the highest-scored 10%, 20% and 30% of people in the later era. Dotted line: chance.

Table 17. Triage value of each score.

Score from	Top 10% of people	Top 20%	Top 30%
Age + sex	30.5%	45.5%	61.0%
+ smoking, BMI	29.5%	50.0%	65.5%
+ 22 routine labs	31.0%	50.0%	65.5%
Whole picture	28.5%	39.5%	46.0%

Share of all five-year cancer deaths found among the highest-scored people in the held-out era. Chance would give 10%, 20% and 30%.

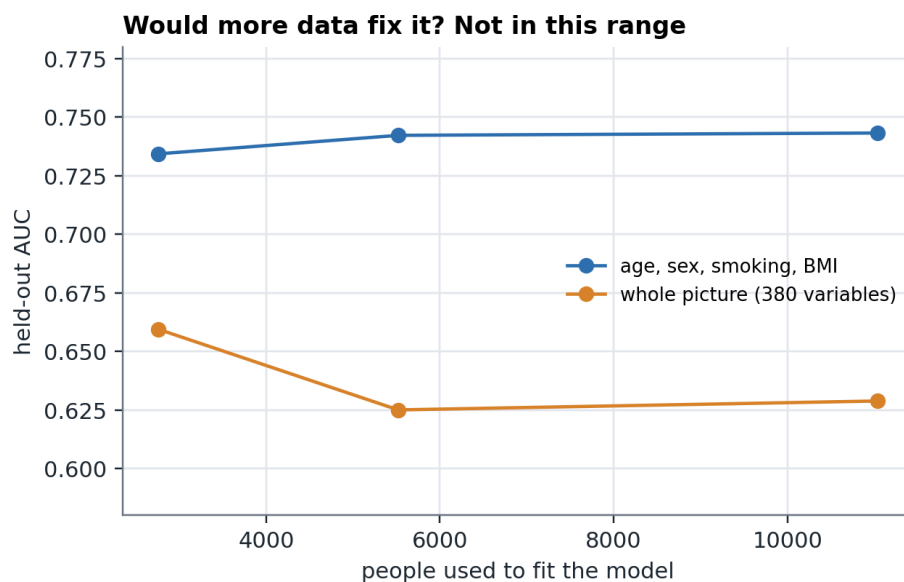


Figure 13. Held-out AUC against the number of people used to fit the model.

Table 18. Does more data help?

People used to fit	Age + sex + smoking + BMI	Whole picture
2,760	0.734	0.659
5,520	0.742	0.625
11,041	0.743	0.629

Held-out AUC. Averaged over three random subsets for the smaller fits.

Table 19. What the whole-picture model relies on.

Variable	Meaning	AUC lost when shuffled
MCQ140__missing	a missing-value flag for this questionnaire item	0.0207
age	age	0.0207
RDQ070__missing	a missing-value flag for this questionnaire item	0.0145
RDQ050__missing	a missing-value flag for this questionnaire item	0.0092
HUQ050__missing	a missing-value flag for this questionnaire item	0.0075
PFQ061K	difficulty with a physical function question	0.0047
LBDHDDSI	HDL cholesterol	0.0045
BMXARMC__missing	a missing-value flag for this questionnaire item	0.0037
OSQ010B__missing	a missing-value flag for this questionnaire item	0.0035
PFQ059__missing	a missing-value flag for this questionnaire item	0.0034
OSQ060__missing	a missing-value flag for this questionnaire item	0.0032
LBXRDW	red cell width	0.0032

Permutation importance of the whole-picture model on a held-out sample. Most of the largest entries are missing-value flags: whether a question was asked at all depends on the survey cycle.

Table 20. Follow-up analyses on the risk-ranking result.

Follow-up (pre-registered addendum)	Result	Reading
F1. Curated 29-variable set vs age + sex + smoking + BMI	AUC 0.752 vs 0.743; gain +0.009 (-0.007 to +0.026); calibration slope 0.987; among decedents +0.004 (-0.023 to +0.030)	Bar missed: calibrated, but no reliable gain
F2a. Predict survey era from the missing-value flags alone	AUC 0.927 using 174 flags	The flags identify the era almost perfectly
F2b. Whole picture, fitted and tested on random halves of all cycles	AUC 0.755 (era split: 0.629; no flags, random split: 0.741)	Consistent with a survey-cycle artefact
F3. Labs-only cancer-specificity OR, 1999-2014 vs NHANES III	0.628 vs 1.184; $z = -3.473$, $p = 0.0005$	The two eras differ; the effect is not stable

Written into docs/RISK_RANKING_PREREG.md before they were run.

3.11 Readability

Patient-facing action and headline lines averaged US grade 6.9, with none over grade 10. The first measurement missed the bar: five action lines exceeded grade 10 and the emergency advice scored grade 16.5 because it was a single long sentence. After rewriting without changing meaning, the emergency advice scored grade 4.8. The remaining hard items in the table are one-word symptom labels, where the formula over-scores.

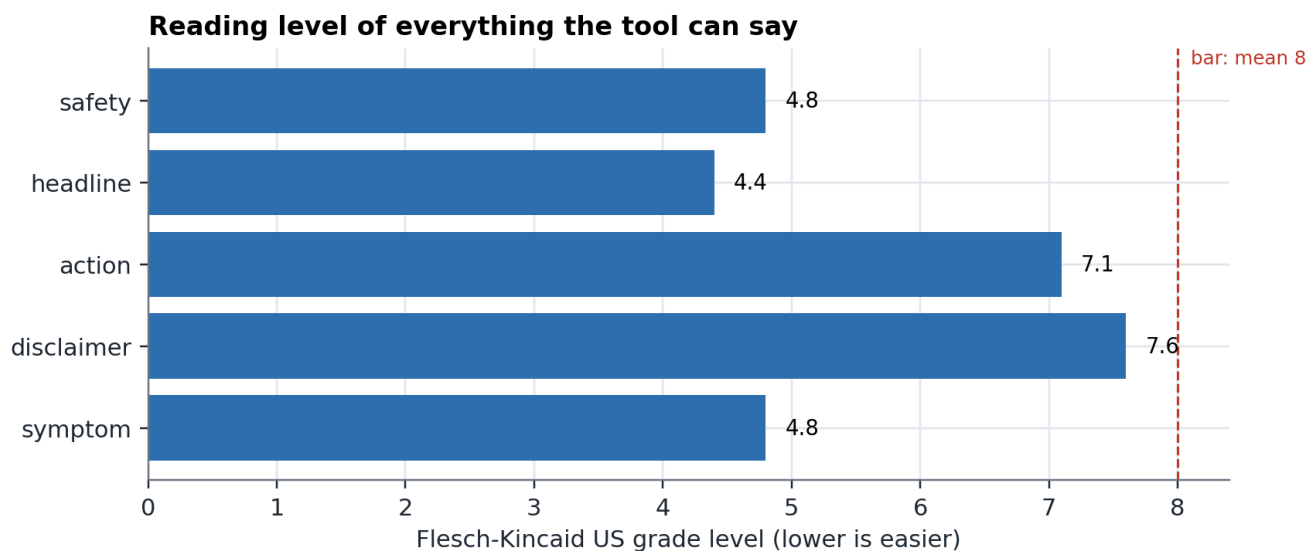


Figure 14. Reading level of each kind of text. Lower is easier; the dashed line is the bar.

Table 21. Reading level of the tool's text.

Text the tool can show	Items	Mean US grade	Hardest item	Over grade 10
Action	34	7.1	9.1	0
Headline	2	4.4	5.8	0
Safety	2	4.8	5.3	0
Disclaimer	1	7.6	7.6	0
Symptom	53	4.8	32.0	5

Flesch-Kincaid grade level. 'Symptom' items are one-line labels, where the formula over-scores very short sentences; five of them exceed grade 10 for that reason only.

3.12 The tool as a person sees it

Figure 15 and Figure 16 are screenshots of the deployed application. The case entered is invented for illustration: a 66-year-old man who has smoked, with unexplained weight loss and a blood count showing iron-deficiency anaemia.

[← Back to Oncovision](#)

Symptom and lab pattern navigator

Tell it what you have noticed and, if you have them, your lab results. It checks whether the combination meets a published guideline threshold for a prompt check, and says what to ask for.

 **RESEARCH PROTOTYPE. NOT MEDICAL ADVICE.**

It applies UK guideline thresholds (the 2015 edition) that a clinician has not yet reviewed for this tool, and it cannot examine you. It never says you have cancer, and meeting no threshold does not mean nothing is wrong. If something worries you, or is getting worse, see a doctor.

Coughing or vomiting a lot of blood, severe chest pain, trouble breathing, or black bloody stools with faintness need urgent care now, not this page.

About you

AGE	SEX	EVER SMOKED	ASBESTOS EXPOSURE
years	Not given ▼	Not given ▼	Not given ▼

What you have noticed

Tick only things with no known cause. If something has an explanation (an injury, a diagnosed condition, a medicine), do not tick it.

CHEST AND BREATHING

☐ **Coughing up blood**
Blood in the phlegm you cough up.

☐ **A cough that has no known cause**

☐ **Shortness of breath with no known cause**

☐ **Chest pain with no known cause**

☐ **Chest infections that keep coming back or will not clear**

Figure 15. The entry form, with the research-prototype warning at the top.

WORTH SEEING A DOCTOR SOON

This combination meets a guideline threshold for a prompt check. Most people who meet it do not have cancer, but it is worth seeing a doctor soon.

Bowel Cancer **SEE A DOCTOR SOON**

Ask your doctor for a specialist appointment within about 2 weeks to check for bowel cancer.

WHY THIS WAS FLAGGED

- aged 60 or over
- iron-deficiency anaemia

Source: NG12 2015, colorectal cancer, p.137

Lung Cancer **SEE A DOCTOR SOON**

Ask for a chest X-ray within about 2 weeks to check for lung cancer.

WHY THIS WAS FLAGGED

- aged 40 or over
- have ever smoked
- any one of: a cough that has no known cause; shortness of breath with no known cause; chest pain with no known cause; weight loss with no known reason; loss of appetite with no known reason; tiredness or fatigue with no known cause

Source: NG12 2015, lung and pleural cancers, p.45

Several Cancers **SEE A DOCTOR SOON**

Weight loss with no known reason can be a sign of several cancers. These include bowel, stomach, gullet, lung, prostate, pancreas, kidney and bladder cancer. It can also have other causes. Guidelines say a doctor should look for the cause soon.

WHY THIS WAS FLAGGED

- weight loss with no known reason

Source: NG12 2015, symptoms that are not specific to one cancer, p.347

Figure 16. An answer for the invented case: what was flagged, why, and where in the guideline it comes from, followed by the age context.

3.13 The screening lookup

The screening lookup has no accuracy to report in the sense of the earlier sections, because it applies written recommendations and predicts nothing. What can be reported is that every age and smoking boundary in the five recommendations is covered by a test written from the published text, and that the page links each answer to its source. Figure 17 shows an invented case: a 58-year-old man who smokes one pack a day and has done so for 30 years. He is told that bowel screening and an annual low-dose CT scan are recommended, that PSA testing is his own choice with his doctor, and is reminded that a family history or gene change could change this.

By US guidelines you may be due for 2 screenings. See below.

Bowel cancer: screening is recommended RECOMMENDED

For ages 50 to 75 the Task Force strongly recommends screening. Choose one of the tests below.

CHOOSE ONE

- a stool test (FIT or a sensitive guaiac test) every year
- a stool DNA test with FIT every 1 to 3 years
- a CT colonography every 5 years
- a flexible sigmoidoscopy every 5 years
- a flexible sigmoidoscopy every 10 years plus a yearly FIT
- a colonoscopy every 10 years

Source: USPSTF, Colorectal Cancer: Screening, May 18 2021 (Grade A for 50 to 75, Grade B for 45 to 49, Grade C for 76 to 85) [read it](#) [🔗](#)

Lung cancer: a low-dose CT scan every year RECOMMENDED

You are in the group the Task Force recommends: aged 50 to 80, at least 20 pack-years, and either smoking now or quit in the last 15 years. Screening stops if your health limits your life expectancy or your ability to have lung surgery.

Source: USPSTF, Lung Cancer: Screening, March 9 2021 (Grade B) [read it](#) [🔗](#)

Prostate cancer: a decision for you and your doctor YOUR CHOICE

For ages 55 to 69 the Task Force says the choice to have a PSA blood test should be yours, after talking about the small possible benefit and the harms, such as false alarms and unnecessary treatment.

Source: USPSTF, Prostate Cancer: Screening, May 8 2018 (Grade C for 55 to 69, Grade D for 70 and older) [read it](#) [🔗](#)

🕒 If a close relative had cancer young, you carry a known gene change, you have had cancer before, or you have had radiation to the chest, ask your doctor. You may need to start sooner or be checked more often.

This is a research prototype and not medical advice. It lists general recommendations for people at average risk. A doctor who knows your history, including any family history of cancer, may advise something different.

Only breast, cervical, colorectal, lung and prostate cancer are covered, because they have a general-population screening recommendation from the Task Force that was checked. Nothing here says other cancers cannot be screened for.

Checked against: The USPSTF recommendation pages on uspreventiveservicestaskforce.org, read on 2026-10-05. NOT YET REVIEWED BY A CLINICIAN.

Figure 17. The screening lookup for an invented 58-year-old man who smokes, with 30 pack-years. Each item links to the Task Force page it comes from.

4. Discussion

4.1 What the results mean

The navigator did what it was designed to do. It applied a published guideline to a person's facts, asked for what was missing, explained each alert with the page it came from, and did so at a reading level most adults can follow. It did not do something it was never designed to do and that a patient-facing tool might be mistaken for: it did not identify who would die of cancer. The NG12 criteria are thresholds for *looking*, set so that the chance of finding cancer is about three percent or more among people who have *already come to a doctor with a symptom*. Applied to symptom-free people using only their lab values, the same thresholds mostly flag the common conditions that cause anaemia and raised white counts.

The decedent-only result needs careful reading. It does not show that anaemia protects against cancer death, and it does not show the guideline is wrong. Anaemia is common in people dying of kidney, heart and lung disease, and in a general-population sample those causes of death are far more common than cancer. Whether an alerted person who then received a stool test or a colonoscopy would have had a bowel cancer found is not something death records can show. What the results do show is that a tool which presents lab-only alerts as a signal of cancer would mislead: the alert lifted the five-year chance of cancer death only from 1.5% to about 3%, mostly by selecting older people, and it marked other causes of death more than cancer.

4.2 The risk-ranking question

The second idea, ranking risk from routine information, produced a clear and useful negative. A four-number model (age, sex, smoking, body mass index) reaches AUC about 0.74 in a later era, is calibrated, and finds roughly half of five-year cancer deaths in its top fifth. Nothing we added, whether 22 routine blood values, 380 variables, or a curated set of 29 clinically chosen ones, reliably improved on it, and giving the larger model four times the data did not help. This does not mean no better model exists. It means that in this survey, with this outcome, the available extra variables carry little cancer-specific signal, and that the apparent gains of a broad model can come from features that identify which survey cycle someone was in. A claim that a broad model detects cancer risk should therefore always show out-of-era performance against a strong baseline, which is what the pre-registration here required.

It is also worth being clear about what a stronger model would need. Stronger discrimination for cancer would most likely come from information this survey does not have: symptoms, family history, prior imaging, and diagnoses rather than deaths as the outcome. Those are the data a clinical-records study would bring, and why clinician collaboration is the next step, not more survey data.

4.3 Changes made because of the results

Lab-only alerts now carry a note that common causes are not cancer and that, in a US survey, these findings did not help tell who later died of cancer once age and sex were known. A lower-tier alert for a cancer site is no longer shown beside a higher-tier one. The context layer was added to the tool because its

bar was met. A rule defect found along the way (asbestos wording shown to a smoker without exposure) was fixed and recorded. These changes are listed as deviations in the pre-registration, which was committed before the first analysis.

4.4 Relation to existing work

QCancer-style models report AUCs of roughly 0.84 to 0.88 for cancer when symptoms, risk factors and blood tests are combined in primary-care records.² Those models use a much richer symptom set and diagnoses as the outcome; the figures are not comparable with an AUC for cancer death in a survey, and we make no claim to match them. Risk-ranking from routine data is therefore an established idea, not a new one. What this work adds is a patient-facing way to read a guideline with a lab report in hand, an honest measurement of what routine survey information does and does not add to age, and a public record of bars set before results. A short search found that QCancer is publicly available online, that Cancer Research UK publishes an interactive NG12 guide for health professionals, and that commercial calculators exist; it found no patient-facing tool that reads a lab report with symptoms and shows the guideline page behind each alert. A short search does not show that none exists, so we do not claim novelty (docs/NOVELTY_AND_REFERENCES_CHECK.md).

4.5 Limitations

- **Not clinician-reviewed.** The rule file records `review_status: not yet reviewed`, and the tool shows this to every user. Transcription errors are possible. A design decision that departs from the guideline text (gating the lung "consider chest X-ray" thrombocytosis path behind at least one symptom, because applied literally it alerted 3.7% of symptom-free adults over 40) needs a clinician's confirmation.
- **Dated and foreign guidance.** The rules are the 2015 edition of UK guidance, which has since been revised (for example, the bowel pathway now uses the faecal immunochemical test). US guidance may differ.
- **Death, not diagnosis.** The outcome cannot show whether an alert would have led to a correct referral. Cancers that were found early and cured are counted as non-cases.
- **Proxies for symptoms.** NHANES has three crude symptom stand-ins and none of the symptoms that drive most of the guideline's high-value rules (bleeding, lumps, bowel change, jaundice). Results for symptom-plus-lab combinations are therefore weak evidence.
- **Few cancer deaths for the risk models.** The fitting era has 179 five-year cancer deaths and the test era 200, so differences of one or two AUC points are uncertain, as the intervals show. Conclusions about the large failure of the whole-picture model rest on a much bigger difference.
- **Self-reported weight and unweighted analysis.** The proxy for weight loss misclassifies people. The analyses ignore survey weights and describe the sample, not the US population.
- **Competing risks.** Non-cancer deaths inside the window are treated as non-cases, which a cause-specific hazard model would handle more carefully.
- **One post hoc analysis in the risk study.** The no-missing-flags variant and the explanation for the failure were added after seeing the result and are labelled as such.

- **Reading level is not comprehension.** A short sentence can still be misunderstood. A protocol for testing it with readers is written (docs/COMPREHENSION_STUDY_PROTOCOL.md) and has not been run.
- **The screening lookup is unreviewed and narrow.** It covers five cancers for average-risk adults, applies guidelines that are revised over time, and has not been reviewed by a clinician.
- **Current NG12 not read at source.** NICE's website refused automated access, so the 2023 changes to the bowel pathway are known only from secondary sources.

4.6 Next steps

1. Clinician sign-off of every rule against the current guideline, recorded in the rule file.
2. A vignette study in which clinicians independently apply the guideline text to written cases and their answers are compared with the tool's, with under-alerting counted separately because it is the harmful direction (docs/NAVIGATOR_VALIDATION_PROTOCOL.md).
3. A comprehension check with lay readers, after an ethics determination.
4. An evaluation on data with real symptoms and diagnoses, such as a clinical records database, once access is obtained: this is where a stronger model could actually be built.
5. Reconciling the rules with current UK and US guidance.

5. Conclusion

A guideline-based navigator can be built, can be made readable, and can be tested against bars set in advance. In this test it applied the guideline as written and produced a stable, modest alert burden. Its lab-driven alerts did not identify who would die of cancer, and a model that used everything a national health survey recorded did no better than age, sex, smoking and body mass index, which together rank cancer death with an AUC of about 0.74. Five of eight pass-or-fail bars were missed and the paper says so. The honest claim is therefore a modest one: the navigator is a faithful, readable way to apply a published referral guideline, accompanied by a calibrated statement of how common cancer death is for a person's age, and it should not be presented as a detector. A stronger model will need symptom and diagnosis data this survey does not have, and whether the navigator helps real people reach a diagnosis sooner is untested.

Declarations

AI assistance

This project was developed with the assistance of an AI model (Anthropic Claude, Sonnet and earlier versions, via Claude Code). The AI was used to write and test code, to run analyses, to draft text, and to search for background literature. An AI cannot be an author. The human author is responsible for the work, reviewed the results, and must verify every statement and reference before submission; the references were looked up by title in Crossref on 5 October 2026 and five of them match exactly; the NICE

guideline, the WHO report, the NCHS mortality files and the 1975 Navy report are not indexed there and must be checked by hand.

Ethics

The analyses use public, de-identified records and involve no contact with participants. Any study recruiting readers or patients will need an ethics determination first.

Data and code availability

Code, rules, tests, pre-registration and all result files are in the public repository (github.com/palashraks-afk/OncoVision). NHANES data are public from the US National Center for Health Statistics; the linked mortality files are public-use. The navigator is deployed at oncovisionai.vercel.app/navigator.

Clinical mentorship and authorship

Clinical mentorship is being sought from Dr. Rishikesh Chavan (UCI Health / CHOC). His role, review of the rules and authorship status are to be agreed with him and are not asserted here.

Conflicts of interest and funding

None declared. No funding.

Warning

The navigator is a research prototype and is not medical advice. A person who is worried about a symptom should see a doctor and should seek urgent care for severe symptoms regardless of what the tool says.

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References 2, 3, 5, 8 and Flesch (in 4) were verified in Crossref. References 1, 6, 7 and Kincaid (in 4) were not machine-verified.

Appendix: the rules

The full rule file, with every criterion and source page, is `backend/navigator_rules.json`. A reviewing clinician can read it directly. Each rule's action text, as the tool shows it to a person, is listed below with its source page; the criteria behind each are in the file.

Rule	Site	Level	What the tool tells the person	Source
lung_haemoptysis	lung cancer	talk soon	Ask your doctor for a specialist appointment within about 2 weeks to check for lung cancer.	NG12 2015, lung and pleural cancers, p.45
lung_xray_symptoms	lung cancer	talk soon	Ask for a chest X-ray within about 2 weeks to check for lung cancer.	NG12 2015, lung and pleural cancers, p.45
lung_xray_consider	lung cancer	worth raising	Ask your doctor if a chest X-ray within about 2 weeks is worth doing.	NG12 2015, lung and pleural cancers, p.45
meso_xray_symptoms	mesothelioma	talk soon	Ask for a chest X-ray within about 2 weeks. Doctors check for mesothelioma when these symptoms come with asbestos exposure.	NG12 2015, lung and pleural cancers, p.49
upper_gi_urgent	oesophageal or stomach cancer	talk soon	Ask for a camera test of the gullet and stomach (an endoscopy) within about 2 weeks.	NG12 2015, oesophageal and stomach cancers, pp.62 and 84
upper_gi_haematemesis	oesophageal or stomach cancer	mention	Tell a doctor. A non-urgent endoscopy may be considered. Vomiting a lot of blood is an emergency: seek care now.	NG12 2015, oesophageal and stomach cancers, pp.62 and 84
upper_gi_consider	oesophageal or stomach cancer	mention	Ask whether a non-urgent endoscopy is worth doing.	NG12 2015, oesophageal and stomach cancers, pp.62 and 84
pancreas_jaundice	pancreatic cancer	talk soon	Ask your doctor for a specialist appointment within about 2 weeks.	NG12 2015, pancreatic cancer, p.71
pancreas_scan	pancreatic cancer	worth raising	Ask whether an urgent scan (CT, or ultrasound if CT is not available) is worth doing.	NG12 2015, pancreatic cancer, p.71
colorectal_urgent	bowel cancer	talk soon	Ask your doctor for a specialist appointment within about 2 weeks to check for bowel cancer.	NG12 2015, colorectal cancer, p.137
colorectal_mass	bowel cancer	worth raising	Ask your doctor if you need a specialist appointment within about 2 weeks.	NG12 2015, colorectal cancer, p.137
colorectal_young_bleeding	bowel cancer	worth raising	Ask your doctor if you need a specialist appointment within about 2 weeks.	NG12 2015, colorectal cancer, p.137
colorectal_stool_test	bowel cancer	worth raising	Ask for a stool test for hidden blood.	NG12 2015, colorectal cancer, p.137

Rule	Site	Level	What the tool tells the person	Source
anal_mass	anal cancer	worth raising	Ask your doctor if you need a specialist appointment within about 2 weeks.	NG12 2015, anal cancer, p.144
breast_urgent	breast cancer	talk soon	A doctor should arrange an urgent breast clinic appointment (within about 2 weeks).	NG12 2015, breast cancer, p.155
breast_consider	breast cancer	worth raising	Ask whether an urgent breast clinic appointment (within about 2 weeks) is needed.	NG12 2015, breast cancer, p.155
breast_young_lump	breast cancer	mention	Ask for a non-urgent referral to have the lump checked.	NG12 2015, breast cancer, p.155
ovarian_symptoms	ovarian cancer	worth raising	Ask for a CA-125 blood test. These symptoms, especially after 50 and on more than about 12 days a month, are the ones doctors test for ovarian cancer.	NG12 (ovarian cancer recommendations dated 2011), p.159
ovarian_ibs	ovarian cancer	worth raising	Ask for tests for ovarian cancer. Irritable bowel syndrome rarely appears for the first time at this age.	NG12 (ovarian cancer recommendations dated 2011), p.159
ovarian_ca125	ovarian cancer	worth raising	A CA-125 at or above 35 with these symptoms means an ultrasound of the abdomen and pelvis should be arranged.	NG12 (ovarian cancer recommendations dated 2011), p.159
ovarian_other	ovarian cancer	worth raising	Ask whether tests for ovarian cancer are worth doing, alongside looking for other causes.	NG12 (ovarian cancer recommendations dated 2011), p.159
prostate_psa	prostate cancer	talk soon	Ask your doctor for a specialist appointment within about 2 weeks.	NG12 2015, prostate cancer, p.178
prostate_symptoms	prostate cancer	mention	Ask about a PSA blood test and a prostate examination.	NG12 2015, prostate cancer, p.178
bladder_urgent	bladder cancer	talk soon	Ask your doctor for a specialist appointment within about 2 weeks.	NG12 2015, bladder cancer, p.189
bladder_uti	bladder cancer	mention	Ask for a non-urgent referral to look for a cause of repeated urine infections.	NG12 2015, bladder cancer, p.189
renal_haematuria	kidney cancer	talk soon	Ask your doctor for a specialist appointment within about 2 weeks.	NG12 2015, renal cancer, p.203
leukaemia_children_blood_count	leukaemia	talk soon	Ask for a blood count within about 2 days. In children and young people these symptoms are the ones doctors check a blood count for, to look for leukaemia.	NG12 2015, leukaemia, p.256

Rule	Site	Level	What the tool tells the person	Source
leukaemia_children_petechiae	leukaemia	talk soon	Tiny red or purple spots that do not fade under pressure in a child or young person need to be seen by a specialist straight away. Seek urgent care now.	NG12 2015, leukaemia, p.256
leukaemia_adult_blood_count	leukaemia	worth raising	Ask whether a blood count within about 2 days is worth doing. Most people with tiredness or similar symptoms do not have leukaemia.	NG12 2015, leukaemia, p.256
lymphoma_adult	lymphoma	worth raising	Ask your doctor if you need a specialist appointment within about 2 weeks. Fever, night sweats, breathlessness, itching, or weight loss make it more likely.	NG12 2015, non-Hodgkin and Hodgkin lymphoma, pp.274 and 280
lymphoma_children	lymphoma	worth raising	Ask whether a specialist appointment within about 2 days is needed. Fever, night sweats, breathlessness, itching, or weight loss make it more likely.	NG12 2015, non-Hodgkin and Hodgkin lymphoma, pp.274 and 280
weight_loss	several cancers	talk soon	Weight loss with no known reason can be a sign of several cancers. These include bowel, stomach, gullet, lung, prostate, pancreas, kidney and bladder cancer. It can also have other causes. Guidelines say a doctor should look for the cause soon.	NG12 2015, symptoms that are not specific to one cancer, p.347
appetite_loss	several cancers	talk soon	Losing your appetite for no known reason can be a sign of several cancers. These include lung, gullet, stomach, bowel, pancreas, bladder and kidney cancer. It can also have other causes. Guidelines say a doctor should look for the cause soon.	NG12 2015, symptoms that are not specific to one cancer, p.347